Multinomial Logistic Regression

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Introduction

Multinomial logistic regression extends binary logistic regression to outcomes with three or more unordered categories. Where binary logistic models the log-odds of one category vs another, multinomial models the log-relative-risk-ratios of each non-reference category against a chosen reference, simultaneously across all $k-1$ contrasts. It is the standard tool for predicting voting choice, transportation mode, disease subtype, or any categorical outcome whose categories have no inherent ordering.

Prerequisites

A working understanding of binary logistic regression, the softmax function, and the relative-risk-ratio interpretation of exponentiated coefficients.

Theory

For outcome $Y \in \{1, \ldots, k\}$ with category 1 as reference:

$$\log \frac{\mathrm{P}(Y=j)}{\mathrm{P}(Y=1)} = \beta_0^{(j)} + \sum_l \beta_l^{(j)} X_l, \qquad j = 2, \ldots, k.$$

The model fits $k-1$ linear equations simultaneously by maximum likelihood. Exponentiating $\beta_l^{(j)}$ gives the relative-risk ratio: the multiplicative change in $\mathrm{P}(Y=j)/\mathrm{P}(Y=1)$ per unit increase in X_l, holding other predictors fixed.

The independence-of-irrelevant-alternatives (IIA) assumption — that the ratio of probabilities between any two categories does not depend on which other categories are available — is the key restriction. When IIA fails, nested or mixed logit alternatives generalise the model.

Assumptions

Unordered categorical outcome (use ordinal logistic for ordered outcomes), independent observations, IIA, and a sufficiently large sample to estimate $k-1$ sets of coefficients (rule of thumb: ≥ 10 events in the smallest category per coefficient).

R Implementation

```r
library(nnet); library(broom); library(gtsummary)

data(iris)
fit <- multinom(Species ~ Sepal.Length + Sepal.Width, data = iris, trace = FALSE)

# Coefficients (log-RRR)
summary(fit)

# Exponentiated as relative-risk ratios
```

```r
tidy(fit, exponentiate = TRUE, conf.int = TRUE)

# Predicted probabilities
head(predict(fit, type = "probs"))

# Publication-ready table
tbl_regression(fit, exponentiate = TRUE)
```

Output & Results

`summary(fit)` returns coefficient tables per non-reference category; `tidy(fit, exponentiate = TRUE)` provides relative-risk ratios with confidence intervals. `predict(..., type = "probs")` returns predicted category probabilities for new observations.

Interpretation

A reporting sentence: "Compared to setosa (reference), each 1-cm increase in sepal length multiplied the relative risk of versicolor over setosa by 10.8 (95 % CI 3.7 to 31.2, $p < 0.001$); the corresponding RRR for virginica was 9.0 (95 % CI 3.1 to 26.0, $p < 0.001$). The reference category (setosa) was chosen because it is one of the three iris species and serves as the baseline group." Always state the reference category and the assumed IIA.

Practical Tips

- Test IIA with the Hausman-McFadden test (`mlogit::hmftest`); if violated, switch to nested or mixed logit (`mlogit`).
- Choose the reference category for the most interpretable contrasts — typically the most common category or a substantively meaningful baseline.
- `nnet::multinom()` uses BFGS optimisation; for very large problems with many categories or predictors, `mclogit` or `mlogit` scale better.
- For ordered outcomes, ordinal logistic regression is more efficient than multinomial; check ordering of levels before defaulting to multinomial.
- Communicate results via predicted probability plots (`ggeffects::ggpredict`); readers absorb absolute probabilities better than relative-risk ratios.
- Variable selection in multinomial models is awkward because each predictor produces $k - 1$ coefficients; consider penalised multinomial logistic (`glmnet(family = "multinomial")`).

R Packages Used

`nnet::multinom()` for the canonical implementation; `mlogit` for choice models with rich panel and nested-logit extensions; `glmnet(family = "multinomial")` for penalised multinomial logistic; `broom::tidy()` for tidy output; `gtsummary::tbl_regression()` for publication-ready tables; `ggeffects` for predicted probability plots.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI